

Numerical Methods Analysis

Sem 3

Watermarked study resource

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CATEGORY: Notes

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===== SOURCE_FILE: sem 3/Numerical Methods & Analysis/PYQs/Formulas (Unit 1 to 5).pdf =====

CATEGORY: PYQs

STATUS: ok

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FORMULA SHEET FOR NUMERICAL METHODS FOR ENGINEERS

Part 1: Modeling, Computers and Error Analysis

Error Definitions

True error:

True percent relative error:

Approximate percent relative error:

Stopping criterion:

Terminate computation when $\epsilon_a < \epsilon_s$ where ϵ_s is the desired percent relative error

Taylor Series

Taylor series expansion

$$\begin{aligned} & () () () () \\ & () \\ & () () \end{aligned}$$

where remainder

$$\begin{aligned} & () () \\ & () \text{ or } () \end{aligned}$$

Error propagation

For n independent variables $x_1, x_2, \dots, x_n$ having errors $\tilde{x}_1, \tilde{x}_2, \dots, \tilde{x}_n$, the error in the function f can be estimated via

$$\begin{aligned} & | \\ & | \tilde{x}_1 | \\ & | \tilde{x}_2 | \\ & | \tilde{x}_n | \end{aligned}$$

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Part 2: Roots of Equations

Method Formulation

Bisection

If () () set

If () () set

False Position ()()

() () If () () set

If () () set

Newton Raphson ()

()

Secant

()()

() ()

Part 3: Linear Algebraic Equations

Gauss Elimination

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LU decomposition

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Decomposition

Back Substitution

Forward Substitution

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Gauss-Seidel method

}

|

|

With relaxation,

()

Part 4: Curve Fitting

Method Formulation Errors

Linear

Regression

where

$$\Sigma(\quad)$$

$$\Sigma \Sigma \Sigma$$

$$\Sigma$$

$$(\Sigma \quad)$$

$$\sqrt{\quad}$$

where:

$$\Sigma(\quad)$$

Polynomial

Regression

For a 2nd order polynomial fit,

where

$$\Sigma(\quad)$$

$$\quad)$$

by differentiating S_r with respect to each coefficients

and setting the partial derivatives equal to zero, we

have:

[

$$\Sigma \Sigma$$

$$\Sigma \Sigma$$

$$\Sigma$$

$$\Sigma$$

$$\Sigma$$

$$\Sigma$$

$$]$$

$$\{$$

$$\}$$

$$\{$$

$$\Sigma$$

$$\Sigma$$

$$\Sigma$$

$$\}$$

$$\sqrt{\quad}$$

$$(\quad)$$

where:

$$\Sigma(\tau)$$

Multiple Linear Regression

For a two-variable linear fit,

where

$$\Sigma ()$$

by differentiating S_r with respect to each coefficients
and setting the partial derivatives equal to zero, we
have:

[

$$\begin{matrix} \Sigma & \Sigma \\ \Sigma & \Sigma \\ \Sigma & \\ \Sigma & \Sigma & \Sigma \end{matrix}$$

]

{

}

{

$$\begin{matrix} \Sigma \\ \Sigma \\ \Sigma \end{matrix} \}$$

$$\sqrt{\quad}$$

where:
 $\Sigma(\tau)$

Newton's
 divided
 difference
 interpolating
 polynomial

For third order:
 $(\quad)(\quad)(\quad)(\quad)$
 $(\quad)(\quad)(\quad)$

where
 $(\quad)$
 $[\quad]$
 $[\quad]$
 $[\quad]$

Lagrange
 interpolating
 polynomial

$$(\quad) \Sigma(\quad)(\quad)$$

where,
 $(\quad) \Pi$

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Part 6: Numerical Differentiation and Integration

A. Numerical Differentiation

Method Formulation Errors

Forward finite-
divided difference

First Derivative:

() () ()

()

() () () ()

()

Backward finite-
divided difference

First Derivative:

() () ()

()

() () () ()

()

Centred finite-
divided difference

First Derivative

() () ()

()

() () () () ()

()

B. Numerical Integration

Method Formulation

Trapezoidal rule () () ()

Multiple-application

trapezoidal rule ()

(() Σ ()

())

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Simpson's 1/3 rule () () () ()

Multiple-application

Simpson's 1/3 rule

()

(() Σ ()

Σ ()

())

Simpson's 3/8 rule () () () () ()

Gauss Quadrature

Gauss-Legendre

() () ()

For two-point Gauss-Legendre:

For three-point Gauss-Legendre:

Change of variables:

() ()

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Part 7: Ordinary Differential Equations

Method Formulation

Euler's First-Order RK

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Heun's Second Order RK (

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()

()

Midpoint Second Order RK

()
(

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Ralston's Second Order RK (

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()
(

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Classical Fourth Order RK

()
()
(

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(

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()

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Part 8: Partial Differential Equations

Method Formulation

Elliptic PDEs

Liebmann's Method

Parabolic PDEs (one
dimensional)

Explicit Method

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Simple Implicit Method

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Crank-Nicolson Method

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()

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